

Advanced programming methodologies for photonic processors

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1 Introduction

Motivation

Talk about limitations of electrical integration circuit and challenges in the post-moore's law era:

- Explosive growth in data transmission demand due to AI, ML..., Electronic processor reaches its limitations, slow moore low, power, bandwidth, I/O bottle neck
- Requirements of new paradigm AI data center: energy efficiency, low latency, interconnect bandwidth, dynamic...

Talk about the state of the art to introduce OCS. These leading industry demonstrate that OCS become a strategic technology choice for overcoming performance and energy efficiency bottlenecks in AI data centers.

- google palomar, spine replacement, dynamic network
- nvidia paper, failure resilience

One of the core technic for OCS is use silicon photonic, talk about photonic processor:

- Photonic integrated circuits are solutions, high bandwidth, low latency, passive, low power consumption
- Optical interconnects are solutions for AI data transmission, especially optical circuit switch, talk about advantages, network reconfiguration

Talk about the importance of photonic hardware programming:

- Photonic integrated circuits they are programmable (PIP through MZI)
- Design and control complexity of the large-scale photonic processor
- Need a high level control layer to manage and configure photonic processor based products in datacenter
- Missing an application oriented programming abstractions for photonic processor design and controlling

Objectives

Research objectives:

- Provide an advanced, efficient, and universal programming methodology for programmable photonic processors in AI data centers
- Resolve how to efficiently map the datacenter applications to the photonic processor
- Address how to comprehensively design and simulate large scale OCS for datacenter
- Explore and develop an SDN of the layer 1 OCS for AI datacenter applications controller

Key contributions:

- Graph-based advanced programming methodologies for photonic applications
- Framework for Optical Circuit Switch Architecture Design
- SDN control system for Optical Circuit Switch

Structure of the thesis

Chapter 2: Theoretical and Programming Methodologies of Photonic Integrated Circuits

- talk about phase shifter / MMI / MZI
- circuits, feedforward / circular
- graphic theory
- sdn theory

Chapter 3: Optical Networking Applications Based on Programmable Integrated Photonic

- interconnect (result)
- beamsplitter (algorithm + result)
- switch (algorithm + result)

Chapter 4: optical circuit switch architectures for datacenter applications.

- type of topologies Benes/Dilated Benes/Piloss/Dilated banyan/Double Dilated Benes
- blocking study packets traffic / ocs traffic

- sax simulations (ILs / SNR)
- VPI link simulation with a STD SOA specification?

Chapter 5: software defined network for OCS

- custom yang model
- netconf
- gnmi

Chapter 6: discussions

- address large-scale ocs (losses / crossing) problem
- ocs datacenter use cases maturity (reconfigurable topologies challenging to implement)
- missing an AI network optimization control software
- sdn standardization process
- conclusions
- future work
 - OpenConfig
 - new architectures
 - multicore approaches